

# A $C^*$ -Annular Diagonal Criterion for Failure of BCP

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June 17, 2026

## Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital  $C^*$ -algebra  $C(K)$  has BCP/UBCP exactly when  $K$  has a countable  $\pi$ -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$  of  $K$  is called a  $\pi$ -basis (weak basis) for  $K$  if every nonempty open subset of  $K$  contains some  $U_\alpha$  in  $\mathcal{U}$  [37, Page 15].

In the setting of commutative  $C^*$ -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

**Corollary 3.9.** *Let  $\mathcal{A}$  be a unital commutative  $C^*$ -algebra. Then the followings are equivalent:*

- (1)  $\mathcal{A}$  has the BCP;
- (2) the pure state space  $\mathfrak{P}(\mathcal{A})$  has a countable  $\pi$ -basis;
- (3)  $\mathcal{A}$  has the UBCP.

In particular, there exist commutative  $C^*$ -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

**Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those  $C^*$ -algebras  $\mathcal{A}$  having BCP/UBCP.*

## Proof Intuition

The quotient-level diagonal criterion says that BCP fails when countably many centers can be tested on local pieces that nearly see their quotient norm, while elements of the ideal vanish on those pieces. In  $C^*$ -algebras the local tests usually have the form

$$b \longmapsto a_k b c_k$$

for contractions  $a_k, c_k$  in a multiplier algebra. The global escaping element is then built by gluing local pieces over disjoint annuli or corners.

This packet records the  $C^*$ -version as a theorem. A second, more concrete phase-annular corollary covers the common situation where the local pieces are right annuli  $g_k$ , the local escape is a scalar phase  $\lambda_k g_k$ , and one can glue bounded scalar choices into a single element of the ambient  $C^*$ -algebra.

## A Localizer Criterion

Let  $B$  be a  $C^*$ -algebra, let  $I \triangleleft B$  be a closed two-sided ideal, and let

$$q : B \longrightarrow B/I$$

be the quotient map. We write  $M(B)$  for the multiplier algebra of  $B$ . If  $B$  is unital,  $M(B) = B$ .

**Definition 1.** *The quotient  $B/I$  has the  $C^*$ -localizer diagonal property if for every sequence  $(b_i)_{i \geq 1} \subset B$  there are:*

- a map  $\pi : \mathbb{N} \rightarrow \mathbb{N}$  with every fiber infinite;
- contractions  $a_k, c_k \in M(B)$ ;
- a null sequence  $\eta_k \downarrow 0$ ;
- elements  $u_k \in B$ ;
- an element  $y \in B$  with  $q(y) \in S_{B/I}$ ;

such that:

1. Assigned near-norming

$$\|a_k b_{\pi(k)} c_k\| \geq \|q(b_{\pi(k)})\| - \eta_k.$$

2. Ideal vanishing along fibers: for every  $d \in I$  and every  $i \geq 1$ ,

$$\lim_{\substack{k \rightarrow \infty \\ \pi(k)=i}} \|a_k d c_k\| = 0.$$

3. Local gluing

$$\|a_k y c_k - u_k\| \leq \eta_k.$$

4. Local escape

$$\|u_k - a_k b_{\pi(k)} c_k\| \geq \|a_k b_{\pi(k)} c_k\| - \eta_k.$$

**Theorem 1.** *If  $B/I$  has the  $C^*$ -localizer diagonal property, then  $B/I$  fails BCP.*

*Proof.* Let  $(\xi_i)_{i \geq 1} \subset B/I$  be an arbitrary countable family of centers. Choose representatives  $b_i \in B$ , so  $q(b_i) = \xi_i$ , and choose the data from Definition 1. Put

$$\zeta = q(y) \in S_{B/I}.$$

Fix  $i \geq 1$  and  $d \in I$ . For  $k$  with  $\pi(k) = i$ , the map

$$L_k : B \rightarrow B, \quad L_k(b) = a_k b c_k$$

is contractive, and hence

$$\|y - b_i - d\| \geq \|a_k(y - b_i - d)c_k\|.$$

Using local gluing, local escape, and assigned near-norming gives

$$\begin{aligned} \|a_k(y - b_i - d)c_k\| &\geq \|u_k - a_k b_i c_k\| - \|a_k y c_k - u_k\| - \|a_k d c_k\| \\ &\geq \|a_k b_i c_k\| - 2\eta_k - \|a_k d c_k\| \\ &\geq \|\xi_i\| - 3\eta_k - \|a_k d c_k\|. \end{aligned}$$

Letting  $k \rightarrow \infty$  through the fiber  $\pi^{-1}(\{i\})$ , and using ideal vanishing, gives

$$\|y - b_i - d\| \geq \|\xi_i\|.$$

Taking the infimum over  $d \in I$ , we obtain

$$\|\zeta - \xi_i\| \geq \|\xi_i\|.$$

Thus the unit vector  $\zeta$  is outside every open ball

$$B(\xi_i, \|\xi_i\|).$$

Since the countable family of centers was arbitrary,  $B/I$  fails BCP.  $\square$

## A Phase-Annular Corollary

The next corollary is often the easiest form to check. It packages the Hahn–Banach phase trick and the auxiliary unit-norm check into the hypotheses.

**Theorem 2** (phase-annular criterion). *Let  $B$  be a unital  $C^*$ -algebra and let  $I \triangleleft B$  be a proper closed two-sided ideal. Suppose that for every sequence  $(b_i)_{i \geq 1} \subset B$ , after setting  $b_0 = 1_B$ , there are:*

- *a map  $\pi : \mathbb{N} \rightarrow \{0, 1, 2, \dots\}$  with every fiber infinite;*
- *contractions  $g_k \in B$ ;*
- *a null sequence  $\eta_k \downarrow 0$ ;*
- *for every scalar sequence  $\lambda = (\lambda_k) \in \ell_\infty$  with  $\|\lambda\|_\infty \leq 1$ , an element  $\Phi(\lambda) \in B$ ;*

*such that:*

1. Assigned right-annulus near-norming

$$\|b_{\pi(k)}g_k\| \geq \|q(b_{\pi(k)})\| - \eta_k, \quad k \geq 1.$$

2. Ideal vanishing: for every  $d \in I$  and every  $i \geq 0$ ,

$$\lim_{\substack{k \rightarrow \infty \\ \pi(k)=i}} \|dg_k\| = 0.$$

3. Scalar gluing

$$\|q(\Phi(\lambda))\| \leq 1, \quad \|\Phi(\lambda)g_k - \lambda_k g_k\| \leq \eta_k \quad (k \geq 1).$$

*Then  $B/I$  fails BCP.*

*Proof.* Let  $(\xi_i)_{i \geq 1} \subset B/I$  be an arbitrary countable family, and choose representatives  $b_i \in B$ . Add  $b_0 = 1_B$  and choose the data in the statement.

For each  $k$  with  $\pi(k) \geq 1$ , choose  $\lambda_k \in \mathbb{T}$  such that

$$\|\lambda_k g_k - b_{\pi(k)}g_k\| \geq \|b_{\pi(k)}g_k\|.$$

Indeed, if  $b_{\pi(k)}g_k \neq 0$ , take a norm-one functional  $\varphi$  with  $\varphi(b_{\pi(k)}g_k) = \|b_{\pi(k)}g_k\|$ , and choose  $\lambda_k \in \mathbb{T}$  so that  $\operatorname{Re}(\lambda_k \varphi(g_k)) \leq 0$ . If  $b_{\pi(k)}g_k = 0$ , any  $\lambda_k \in \mathbb{T}$  works. For  $k$  with  $\pi(k) = 0$ , put  $\lambda_k = 1$ .

Let

$$y = \Phi((\lambda_k)_{k \geq 1}).$$

The scalar gluing hypothesis gives  $\|q(y)\| \leq 1$ . We claim that  $\|q(y)\| = 1$ . Fix  $d \in I$  and let  $k \rightarrow \infty$  through the fiber  $\pi^{-1}(\{0\})$ . Since  $b_0 = 1_B$ ,

$$\|g_k\| = \|b_0 g_k\| \geq \|q(1_B)\| - \eta_k = 1 - \eta_k.$$

Therefore

$$\|y - d\| \geq \|(y - d)g_k\| \geq \|\lambda_k g_k\| - \|y g_k - \lambda_k g_k\| - \|d g_k\| \geq 1 - 2\eta_k - \|d g_k\|.$$

Letting  $k \rightarrow \infty$  along that fiber gives  $\|y - d\| \geq 1$ , and then taking the infimum over  $d \in I$  gives  $\|q(y)\| \geq 1$ . Hence

$$q(y) \in S_{B/I}.$$

Now fix  $i \geq 1$  and  $d \in I$ . Let  $k \rightarrow \infty$  through  $\pi^{-1}(\{i\})$ . Then

$$\begin{aligned} \|y - b_i - d\| &\geq \|(y - b_i - d)g_k\| \\ &\geq \|\lambda_k g_k - b_i g_k\| - \|y g_k - \lambda_k g_k\| - \|d g_k\| \\ &\geq \|b_i g_k\| - \eta_k - \|d g_k\| \\ &\geq \|q(b_i)\| - 2\eta_k - \|d g_k\|. \end{aligned}$$

Letting  $k \rightarrow \infty$  along the fiber and then taking the infimum over  $d \in I$ , we obtain

$$\|q(y) - \xi_i\| \geq \|\xi_i\|.$$

Thus  $q(y)$  is a unit vector outside every proposed ball  $B(\xi_i, \|\xi_i\|)$ , and  $B/I$  fails BCP.  $\square$

## Recovering Existing Diagonal Packets

### Sigma-Unital Multiplier Coronas

Let  $A$  be a nonunital  $\sigma$ -unital  $C^*$ -algebra. In the sigma-unital corona packet, one takes

$$B = M(A), \quad I = A.$$

An increasing positive contractive approximate unit  $(e_n)$  with  $e_m e_n = e_n$  for  $m \geq n$  supplies inner annuli

$$g_k = e_{m_k} - e_{n_k}$$

and larger outer annuli

$$h_k = e_{b_k} - e_{a_k}, \quad h_k g_k = g_k, \quad h_j g_k = 0 \quad (j \neq k).$$

Finite annulus recovery gives

$$\|T_{\pi(k)} g_k\| \approx \|q(T_{\pi(k)})\|,$$

and  $A$ -elements vanish on far-out annuli. For  $\lambda \in \ell_\infty$ , strict convergence gives the scalar gluing element

$$\Phi(\lambda) = \sum_k \lambda_k h_k \in M(A),$$

with  $\|\Phi(\lambda)\| \leq \|\lambda\|_\infty$  and  $\Phi(\lambda)g_k = \lambda_k g_k$ . Thus Theorem 2 is exactly the  $C^*$ -annular checklist behind the sigma-unital corona theorem.

## Hilbert-Module Coronas

Let  $E$  be a Hilbert  $C^*$ -module with a BCP block system. In the Hilbert-module block packet, one takes

$$B = \mathcal{L}(E), \quad I = \mathcal{K}(E).$$

The localizers in Theorem 1 are

$$a_k = 1, \quad c_k = e_{F_k},$$

where  $e_{F_k}$  is a fresh compact block projection. Tail norm recovery gives assigned near-norming, compact-vanishing gives ideal vanishing, fresh range escape supplies the local pieces  $u_k = v_k$ , and strict gluing gives a single operator  $y = V$  with  $Ve_{F_k} = v_k$ . Hence the Hilbert-module block criterion is an instance of the localizer theorem.

## Ordinary Product Quotients

For an ordinary  $c_0$ -product quotient

$$\prod_n A_n / \bigoplus_{c_0} A_n$$

with unital  $A_n$ 's, disjoint infinite coordinate sets  $T_k$  give central projections  $g_k = 1_{T_k}$ . Elements of the  $c_0$ -sum vanish on these sets as  $k \rightarrow \infty$ , after the sets are chosen pairwise disjoint. Local escape elements can be glued coordinatewise. This is the coordinate version of the same annular principle. For more general ideals  $\mathcal{I}$ , the full reduced-product packet is better viewed through the broader quotient-level asymptotic criterion, because local restrictions naturally land in local reduced quotients rather than in a single ambient  $C^*$ -algebra.

## Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with annular diagonal,  $C^*$ -localizer, multiplier corona, Hilbert-module block criterion, asymptotic diagonal, and quotient diagonal keywords. Existing packets contained the Banach quotient criterion and concrete  $C^*$ -diagonal proofs, but no standalone  $C^*$ -localizer/phase-annular criterion.

The proof of Theorem 1 is a direct quotient-norm diagonal estimate. The phase-annular theorem is a self-contained specialization using the Hahn–Banach phase choice and an auxiliary  $b_0 = 1$  fiber to force the glued quotient element to have norm one.

## Human Review Focus

Check the localizer hypotheses in Definition 1 and the scalar gluing hypothesis in Theorem 2. These are the reusable  $C^*$ -checklists; once they are verified in an application, the BCP failure proof is formal.

## References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.